Advanced ECG Analysis and Deep Learning on PTB-XL

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Abstract

This report investigates automated electrocardiogram (ECG) analysis on the PTB-XL database using progressively richer representation-learning strategies. In Part I, we characterize the dataset, define the diagnostic classification task, and establish supervised baselines using compact convolutional and residual one-dimensional neural networks. The study emphasizes rigorous split management, pathology-oriented exploratory analysis, and benchmark-aware comparison with the PTB-XL literature. Parts II and III will extend this baseline with cross-lead representation learning through a multi-lead autoencoder and a PointNet-inspired classifier operating on ECG-embedding structures.

1 Introduction

Electrocardiography (ECG) remains one of the most widely used non-invasive tools for the detection of cardiac abnormalities. However, manual ECG interpretation is time-consuming, subject to inter-observer variability, and particularly challenging in multi-label clinical settings where several pathologies may co-occur within the same recording [1, 2]. These limitations have motivated a large body of work on deep learning for automated ECG interpretation.

Among publicly available datasets, PTB-XL has emerged as one of the most important benchmarks for reproducible ECG analysis. It provides clinical 12-lead recordings with expert annotations, recommended split information, and a diagnostic hierarchy that supports both fine-grained and coarse-grained classification tasks [2, 3]. The dataset is therefore particularly suitable for comparing deep learning architectures under controlled experimental conditions.

This report follows the structure of the assessment and develops the study in three stages. In Part I, we perform dataset exploration, justify the pathology focus, and establish supervised baselines. In Part II, we train a multi-lead autoencoder that reconstructs all 12 leads from a single-lead input and produces aggregated embeddings. In Part III, we build a PointNet-inspired classifier on a joint ECG-plus-embedding representation. The present manuscript starts with Part I only, with the objective of creating a rigorous benchmark-aware foundation for the later representation-learning stages.

A key methodological principle in this work is *comparability*. Many PTB-XL studies differ in label granularity, preprocessing, signal resolution, and evaluation metrics, which makes naive comparisons misleading. For this reason, our baseline discussion emphasizes results that are either directly comparable to the standard five-superclass PTB-XL task or clearly identified as contextual rather than strictly equivalent references [1, 4, 5, 6].

2 Dataset and pathology focus

We used the PTB-XL PhysioNet release (v1.0.3), a large-scale public 12-lead ECG dataset containing 21,799 clinical recordings from 18,869 patients. Each record is 10 seconds long and is provided both at 500 Hz and in a downsampled 100 Hz version. The annotations follow the SCP-ECG standard and cover diagnostic, rhythm, and form statements; for diagnostic classification, the labels can be aggregated into five clinically meaningful superclasses: *NORM*, *MI*, *STTC*, *CD*, and *HYP*. PTB-XL is inherently a multi-label benchmark, since a single ECG may contain several co-occurring diagnostic statements [2, 3]. In the original PTB-XL publication, ages of 90 years or older are encoded as 300 to preserve anonymity [2].

To ensure benchmark comparability, we followed the official PTB-XL fold assignment and used folds 1–8 for training, fold 9 for validation, and fold 10 for testing, consistent with the recommended patient-wise evaluation protocol [1, 3]. In our cleaned metadata, this resulted in 17,418 training ECGs, 2,183 validation ECGs, and 2,198 test ECGs, with zero patient overlap across partitions. Signals were loaded from the 100 Hz recordings (`filename_lr`), truncated or padded to 1000 samples, and normalized independently for each lead using z-score normalization. These preprocessing and split choices match the actual implementation used in the project data pipeline.

Table 1 summarizes the prevalence of the five diagnostic superclasses in the processed metadata. The dataset is clearly imbalanced: *NORM* is the dominant class, whereas *HYP* is substantially less frequent. At the same time, the task is not a standard single-label problem. In our exploratory analysis, ECGs contained up to four co-occurring diagnostic superclasses, with 21,388 records exhibiting at least one diagnostic superclass and an average of 1.27 superclasses per ECG. This label structure supports treating PTB-XL as a structured multi-label classification problem rather than as a mutually exclusive diagnostic task.

Table 1: Diagnostic superclass prevalence in the PTB-XL metadata used in this work. Counts and prevalences are computed from the processed dataset used in the experiments.

Superclass	Count	Prevalence (%)
NORM	9514	43.64
MI	5469	25.09
STTC	5235	24.01
CD	4898	22.47
HYP	2649	12.15

Among pathological classes, *MI*, *STTC*, and *CD* were selected as the main groups of interest for subsequent analysis. This choice was motivated by two complementary considerations. First, these three superclasses are the most prevalent pathological groups in the dataset, which makes them statistically more suitable for robust supervised learning than *HYP*. Second, they correspond to distinct and clinically relevant ECG phenomena: infarction-related abnormalities for *MI*, repolarization changes for *STTC*, and conduction abnormalities for *CD*. By contrast, *NORM* remains necessary in the predictive task for benchmark comparability, but it is less informative from a pathology-centered perspective.

This pathology focus does not change the supervised setup used for the baseline experiments. To remain directly comparable with prior PTB-XL benchmarks, the classification models in Part 1 were trained on the standard five-superclass task rather than on a reduced three-class subset. This distinction is important: *MI*, *STTC*, and *CD* are emphasized for interpretation and clinical discussion, whereas *NORM* and *HYP* are retained in the prediction space to preserve fidelity to the benchmark literature [1]. The PTB-XL benchmark study by Strodthoff et al.

also emphasizes that strict split design and task comparability are essential, since differences in train–test partitioning, label selection, and evaluation procedures can substantially alter reported performance [1].

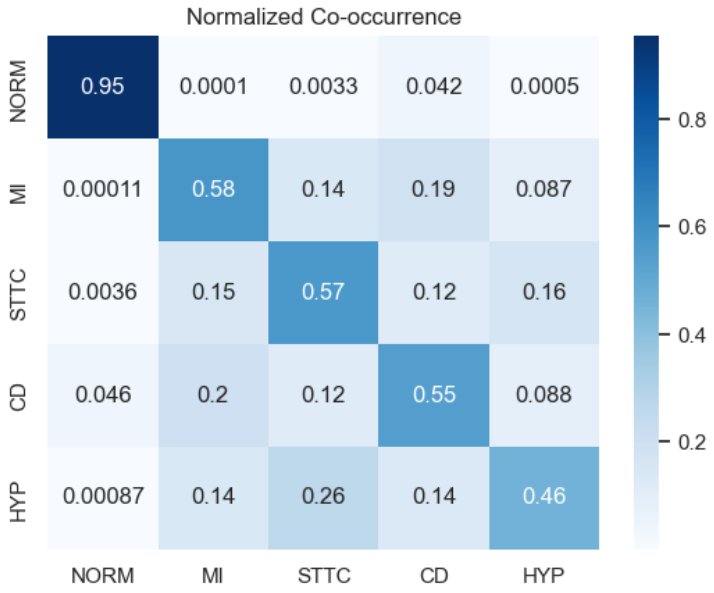


Figure 1: Normalized co-occurrence matrix of the five diagnostic superclasses in PTB-XL. Strong overlap between MI, STTC, and CD supports the multi-label formulation and motivates representation-learning approaches that can capture inter-class dependencies.

3 Baseline models and benchmark context

To establish a reproducible reference for the later representation-learning stages, we implemented two supervised baselines directly on the raw 12-lead ECG signals: a compact CNN1D and a custom ResNet1D. Both models operate on the 100 Hz PTB-XL recordings (`filename_lr`) and receive signals resized to a fixed length of 1000 samples and normalized independently for each lead using z-score normalization. This setup follows a lightweight raw-signal pipeline and remains compatible with the standard PTB-XL benchmarking regime at 100 Hz [1].

The CNN1D baseline was designed as a simple reference architecture. It consists of three one-dimensional convolutional blocks with channel widths 32, 64, and 128 and kernel sizes 7, 5, and 3, with max-pooling after the first two convolutions, followed by global average pooling and a two-layer fully connected classifier. By contrast, the ResNet1D baseline follows a residual design inspired by ResNet-18, with an initial convolutional stem and four residual stages of depths $[2, 2, 2, 2]$ and channel progression 64, 128, 256, and 512. Each residual block contains two convolution–batch-normalization layers and an identity or projection shortcut. In this study, the CNN1D serves as a minimal baseline, whereas the ResNet1D provides a stronger architecture for assessing the benefit of deeper temporal feature extraction.

Training was performed in a multi-label setting using binary cross-entropy with logits. To mitigate class imbalance, the loss was weighted with class-dependent positive weights computed from the training split. Optimization used Adam, and the best checkpoint was selected according to validation macro-AUC. Final test predictions were thresholded at 0.5 to compute class-wise precision, recall, and F1-score. These models are not intended as state-of-the-art reproductions; rather, they define controlled internal baselines against which the autoencoder-derived embeddings and the PointNet-inspired classifier can later be assessed.

Table 2: Benchmark-aware comparison on PTB-XL. Only the row from Strodthoff et al. corresponds to the canonical official benchmark setting; the remaining literature rows are included as closely related context.

Study	Model / setup	Metric	Value
Strodthoff et al. [1]	Official PTB-XL superdiagnostic benchmark, `resnet1d_wang`	Macro-AUC	0.930
Śmigiel et al. [4]	PTB-XL 5-class CNN with entropy features	Accuracy	76.5%
Feyisa et al. [5]	Multi-receptive-field CNN on PTB-XL 5 superclasses	AUC / Macro-F1	0.93 / 0.72
Geng et al. [6]	Multi-task SE-ResNet + CoT attention + Bi-GRU on PTB-XL	Average AUC / F1	0.925 / 0.833
This work	CNN1D baseline	Macro-AUC / Macro-F1	0.880 / 0.635
This work	ResNet1D baseline	Macro-AUC / Macro-F1	0.873 / 0.646

Table 2 places our baselines in the context of prior PTB-XL studies. The most direct comparison remains the official benchmark of Strodthoff et al., who evaluated convolutional, recurrent, and residual architectures on the recommended PTB-XL tasks and reported a macro-AUC of 0.930 for `resnet1d_wang` on the five-superclass diagnostic benchmark [1]. Other studies are informative but should be interpreted more cautiously, since some emphasize accuracy rather than macro-AUC, while others introduce alternative architectures or auxiliary-task formulations [4, 5, 6].

Our results remain below the strongest PTB-XL references, which is expected given the deliberately lightweight scope of the present baselines. This gap is scientifically useful rather than problematic: it establishes a transparent reference point for Parts 2 and 3, where richer learned representations are introduced. Interestingly, the compact CNN1D achieved a slightly higher macro-AUC than the current ResNet1D implementation, whereas the ResNet1D obtained a slightly higher macro-F1 after thresholding. This difference suggests that the CNN1D produced better probabilistic ranking, while the ResNet1D yielded marginally better hard decisions at the chosen threshold.

Finally, we note that recent public work by Kanos explores a broader family of PTB-XL architectures, including CNN, ResNet1D, xResNet1D, Transformer, and xLSTM variants, through both a thesis manuscript and an accompanying repository [7, 8]. However, that study uses standardized 500 Hz waveform data and a different class formulation based on binarized SCP-code targets rather than the exact five-superclass 100 Hz setup adopted here. For this reason, it is cited as contextual implementation evidence rather than as a direct benchmark comparator.

4 Multi-Lead ECG Autoencoder with Embedding Aggregation

4.1 Architecture

To model cross-lead dependencies, we implemented a one-dimensional U-Net autoencoder that reconstructs the full 12-lead ECG from a single observed lead. The network follows an encoder–decoder design with skip connections, which is well suited to ECG reconstruction because it

combines local waveform modeling with multi-scale temporal context. The encoder contains four downsampling stages with channel widths 32, 64, 128, and 256, followed by a bottleneck block with 512 channels. Each block consists of two 1D convolutions with kernel size 3, batch normalization, and ReLU activation. The decoder mirrors this hierarchy with four upsampling stages and skip connections from the encoder.

At the output of the decoder, the feature map has 32 channels. A 1×1 convolution projects this representation into $12 \times 32 = 384$ channels, which are interpreted as 12 lead-specific feature groups of dimension 32. A grouped 1×1 convolution then converts these grouped features into the 12 reconstructed ECG leads. In parallel, the same grouped latent tensor is reshaped into $(12, 32, T)$ and averaged over time to produce 12 lead-wise embeddings of size 32. Concatenating these embeddings yields a unified 384-dimensional representation for each ECG, as required in Part 2 of the assessment.

This design was motivated by two considerations. First, U-Net-style reconstruction architectures have recently shown strong performance in ECG lead reconstruction tasks [9, 10]. Second, coupling reconstruction and embedding extraction within the same network provides a natural way to learn features that may later support downstream classification.

4.2 Training strategy

The autoencoder was trained on the PTB-XL 100 Hz recordings using the same patient-wise train/validation/test split as in Part 1. For each training example, a single input lead was selected and the network was trained to reconstruct the full 12-lead ECG using a mean squared error (MSE) loss:

$$\mathcal{L}_{\text{recon}} = \frac{1}{N} \sum_{i=1}^{N} (\hat{x}_i - x_i)^2.$$

We monitored reconstruction quality using MSE and mean Pearson correlation across leads. Training used Adam with learning rate 10^{-3} for 15 epochs. In addition to the main random-input-lead model, we retained the earlier lead-I-only model as a controlled comparison baseline. This allows a direct comparison between a lead-specialized model and a lead-agnostic model.

4.3 Experimental comparison: fixed-lead vs. random-lead training

The comparison between the two training regimes revealed a clear trade-off. The lead-I-only model achieved better reconstruction when evaluated under the same lead-I conditioning used during training, whereas the random-input-lead variant was substantially less sensitive to the identity of the source lead. This distinction is important for the goals of the assessment: the former behaves as a specialized reconstruction model, while the latter better matches the requirement of choosing one lead at a time and learning a more general cross-lead mapping.

Table 3 summarizes the main results. When evaluated with lead I as input, the lead-I-only model achieved a lower test MSE (0.3952) and higher mean correlation (0.7640) than the random-input-lead variant (0.4704 and 0.7014, respectively). However, when reconstruction was tested across all 12 possible input leads, the lead-I-only model showed very high variability, with MSE values ranging from 0.3952 to 1.6516. By contrast, the random-input-lead model was markedly more stable, with input-lead MSE values confined to a much narrower interval from 0.4421 to 0.5254. This result is consistent with recent arbitrary single-lead reconstruction studies, which report that training with variable source leads improves flexibility and robustness, even if a fixed-lead model may retain an advantage under perfectly matched test conditions [10].

Table 3: Comparison between the lead-I-only and random-input-lead U-Net variants. The fixed-lead model performs best when evaluated under the same lead-I conditioning regime used during training, whereas the random-input model is markedly more robust to the choice of source lead.

Metric	Lead-I only	Random input lead
Best validation MSE	0.3867	0.4733
Best validation correlation	0.7672	0.6990
Test MSE	0.3952	0.4704
Test correlation	0.7640	0.7014
Mean input-lead MSE (12 leads)	0.9372	0.4827
Input-lead MSE range	0.3952–1.6516	0.4421–0.5254
Output lead 0 MSE	0.0045	0.1482

This comparison provides an intuitive interpretation of the two models. The near-perfect reconstruction of output lead 0 by the lead-I-only model should not be interpreted as purely cross-lead inference, because the source and one of the targets coincide. By contrast, the random-input-lead model sacrifices some of this lead-specific specialization in exchange for better generalization across the full set of source leads. From the perspective of downstream representation learning, this second property is more desirable, because it suggests that the learned embeddings are less tied to a single acquisition perspective.

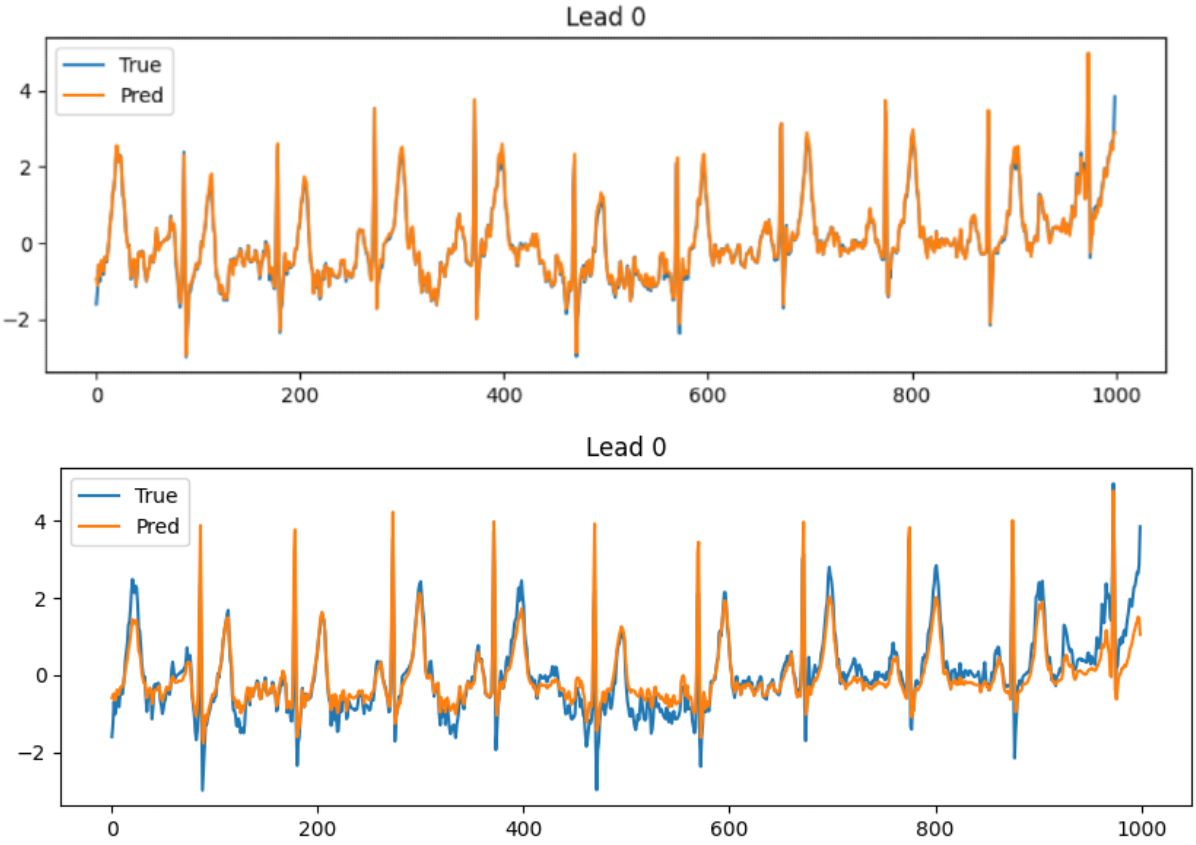


Figure 2: Representative reconstruction of lead 0 under the two training regimes. Top: lead-I-only model. Bottom: random-input-lead model. The fixed-lead model reconstructs lead 0 almost perfectly because the evaluation condition matches the training condition. The random-input model yields a slightly less exact lead-0 reconstruction but is designed to generalize across all possible source leads.

4.4 Lead-wise reconstruction analysis

Lead-wise reconstruction analysis confirms that not all leads are equally easy to reconstruct from a single source lead. For the lead-I-only model, the hardest reconstructed outputs were leads 5 and 2, while for the random-input-lead model they remained among the most difficult outputs. This suggests that some leads contain morphological information that is less directly inferable from a single observed perspective. Such lead-dependent reconstruction differences have also been emphasized in recent work on ECG lead informativeness and lead reconstruction, where performance was shown to depend strongly on the inter-lead information structure [9].

Qualitative inspection of reconstructed signals showed that both variants preserved global rhythm, beat timing, and much of the broad waveform morphology. The largest errors occurred around sharp deflections, amplitude extremes, and some boundary effects near the segment edges. These observations are consistent with the general behavior of signal-reconstruction autoencoders, which often capture coarse morphology well while smoothing high-frequency or high-amplitude details.

A full dynamic time warping (DTW) evaluation was implemented, but the current naive CPU implementation was too slow to complete efficiently on the entire test set.

4.5 Embedding extraction and latent-space analysis

A central goal of Part 2 is not only reconstruction, but also the extraction of informative latent representations. On the test set, the autoencoder produced one 32-dimensional embedding per lead, which were concatenated into a 384-dimensional vector for each ECG. This yielded an embedding matrix of shape $(2198, 384)$, exactly matching the expected dimensionality of 12 lead-wise embeddings of size 32.

To analyze the embedding space, we standardized the embeddings, reduced them to 50 dimensions with PCA, and then visualized them with both t-SNE and UMAP. These methods are widely used for qualitative analysis of high-dimensional representations: t-SNE emphasizes local neighborhood structure, whereas UMAP often provides a more stable view of broader manifold organization [11, 12]. In this report, both are used strictly as exploratory visualization tools rather than as quantitative proof of class separability.

The most robust finding from these visualizations is that the learned embeddings are not random. In both t-SNE and UMAP projections, normal ECGs and pathological ECGs occupy partially distinct regions of the latent space. This indicates that the reconstruction objective captures clinically meaningful variation even without an explicit classification loss. Such behavior is consistent with prior ECG representation-learning studies showing that self-supervised or generative objectives can produce latent representations that remain informative for downstream clinical tasks [13].

At the superclass level, the latent structure is more diffuse. The normal class appears comparatively concentrated, whereas the pathological superclasses remain broader and overlapping. This is expected for two reasons. First, PTB-XL is inherently multi-label, with frequent co-occurrence of abnormalities. Second, the autoencoder is optimized to reconstruct signals, not to maximize discriminative separation between diagnoses. Accordingly, the embedding plots should be interpreted as evidence of latent organization and enrichment, rather than as disease-specific clusters.

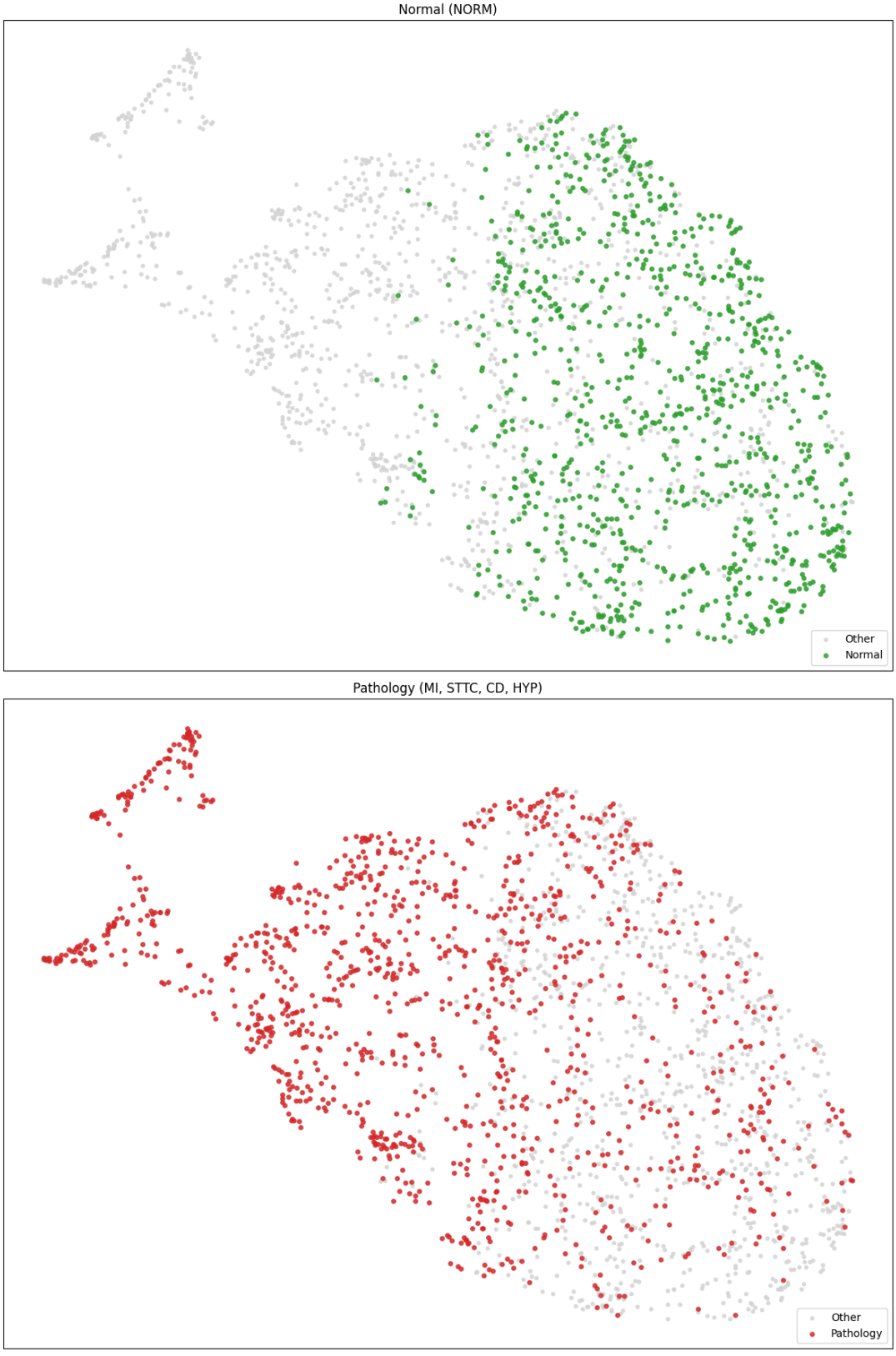


Figure 3: UMAP projection of the 384-dimensional concatenated embeddings extracted from the random-input-lead autoencoder. Normal ECGs and pathological ECGs occupy partially distinct regions of the latent space, indicating that the reconstruction objective captures clinically meaningful variation even without explicit diagnostic supervision.

4.6 Discussion

Overall, the U-Net autoencoder learned non-trivial cross-lead dependencies and produced a structured latent representation suitable for downstream use. The comparison between the lead-I-only and random-input-lead variants highlights a clear methodological trade-off: specialization to a fixed input lead improves performance under matched conditions, whereas random source-lead training substantially improves robustness to lead choice. Because the downstream Part 3 model should ideally benefit from embeddings that are less dependent on a single acquisition perspective, the random-input-lead variant is the more relevant representation-learning model.

At the same time, the present reconstruction results should not be overstated. Recent studies have stressed that single-lead-to-12-lead reconstruction may smooth waveform details or regress toward average morphology, particularly when subtle or strongly patient-specific features must be recovered [10, 9]. Our qualitative examples suggest that the model captures broad rhythm and morphology well, but remains less precise for sharp peaks and certain harder-to-infer leads. For this reason, the autoencoder is best interpreted here as a representation-learning module that encodes clinically meaningful structure, rather than as a clinically deployable ECG reconstruction system.

5 PointNet-Inspired Classification on ECG Point Sets

5.1 Representation strategy

The objective of Part 3 was to adapt a PointNet-style classifier to a structured ECG representation combining the original signal and the latent descriptors learned in Part 2. The central idea was to reinterpret the 12-lead ECG as a set of spatiotemporal points rather than as a conventional matrix or sequence.

Let the preprocessed ECG be denoted by $x \in \mathbb{R}^{12\times T}$, with $T = 1000$ time samples after padding/truncation. Let $\mathbf{e}_\ell \in \mathbb{R}^{32}$ denote the embedding associated with lead ℓ, extracted from the pretrained U-Net autoencoder. For each pair (ℓ, t), corresponding to lead $\ell \in \{1, \ldots, 12\}$ and time index $t \in \{1, \ldots, T\}$, we define one point

$$\mathbf{p}_{\ell,t} = [\, t,\ \ell,\ x_{\ell,t},\ \mathbf{e}_\ell \,] \in \mathbb{R}^{35}.$$

Thus, each ECG is represented as a set of $12 \times 1000 = 12{,}000$ points, and each point has 35 features: time index, lead index, voltage amplitude, and the 32-dimensional embedding of the corresponding lead.

This transformation is implemented in `create_point_cloud()`, which explicitly constructs the time grid and lead grid, keeps the original signal amplitude as the voltage coordinate, and repeats the lead-specific embedding across all time samples of that lead before concatenation. As a result, the representation is best interpreted as an *augmented point set*: the tuple $(t, \ell, x_{\ell,t})$ acts as a structured ECG coordinate, while the latent embedding acts as an attached feature vector. This directly answers the requirement of combining a 2D ECG representation with the embeddings learned in Part 2.

5.2 PointNet-inspired architecture

The classifier used in this work is a simplified PointNet-inspired model. Given an input tensor of shape (B, N, d), where B is the batch size, $N = 12{,}000$ is the number of points per ECG, and $d = 35$ is the point dimensionality, the network applies a shared multilayer perceptron independently to every point:

$$35 \to 64 \to 128 \to 256.$$

The first two linear layers are followed by ReLU activations. The point-wise features are then aggregated by global max pooling across the point dimension, producing a single 256-dimensional descriptor for the entire ECG. A final classifier head

$$256 \rightarrow 128 \rightarrow 5$$

with ReLU and dropout ($p = 0.3$) outputs one logit per diagnostic superclass. This architecture preserves the main design principle of PointNet, namely the combination of shared point-wise feature extraction with a symmetric pooling operation that yields a global set representation. However, the present implementation remains deliberately simple. It does not include learned input-alignment or feature-alignment modules, and it does not model explicit local neighborhoods between points. It should therefore be described as *PointNet-inspired* rather than as a full reproduction of PointNet.

5.3 Training procedure

The U-Net autoencoder from Part 2 was reused as a frozen embedding extractor. During PointNet training, validation, and testing, one source lead was sampled randomly for each ECG, and this single input lead was passed through the U-Net to produce the 12 lead-wise embeddings. These embeddings were then combined with the full original 12-lead ECG through the point-set construction described above. Therefore, the PointNet classifier was trained and evaluated with *random-lead-conditioned embeddings*, rather than embeddings tied to a fixed reference lead.

The classifier was trained for 10 epochs using Adam with learning rate 10^{-3} and `BCEWithLogitsLoss`, consistent with the multi-label nature of the five-superclass PTB-XL task. Validation loss was monitored after each epoch, and it decreased from 0.4402 at epoch 1 to 0.3774 at epoch 10. No class-dependent positive weights, early stopping, or model selection based on validation AUROC were used in this stage. These choices kept the pipeline simple, but they also likely limited performance relative to the stronger Part 1 baselines.

5.4 Classification results

On the test set, the PointNet-inspired classifier achieved a macro-AUC of 0.8286. The corresponding macro-averaged precision, recall, and F1-score were 0.7096, 0.4645, and 0.5328, respectively. Class-wise performance was heterogeneous: the best results were obtained for the normal class, with precision 0.74, recall 0.86, and F1-score 0.80, whereas the pathological classes were more difficult, especially hypertrophy, which reached a recall of only 0.15.

Table 4: Test-set results for the PointNet-inspired ECG classifier.

Class	Precision	Recall	F1-score	Support
CD	0.71	0.48	0.57	496
HYP	0.67	0.15	0.24	262
MI	0.67	0.39	0.49	550
NORM	0.74	0.86	0.80	963
STTC	0.75	0.45	0.56	521
Macro average	0.71	0.46	0.53	–

For clarity, Table 5 compares the PointNet-inspired model with the two supervised baselines from Part 1. Although the PointNet approach is conceptually interesting, it did not outperform the simpler 1D signal classifiers. Its macro-AUC remained below both CNN1D and ResNet1D,

and its macro-F1 was substantially lower. The most honest interpretation is therefore that the proposed 3D point-set representation is feasible and informative, but does not yet provide a meaningful improvement in classification performance.

Table 5: Comparison between the Part 1 baselines and the PointNet-inspired classifier.

Model	Macro-AUC	Macro-F1
CNN1D baseline	0.8799	0.6352
ResNet1D baseline	0.8726	0.6457
PointNet-inspired ECG classifier	0.8286	0.5328

5.5 Methodological interpretation

The main contribution of this part lies in the representation itself rather than in the raw classification score. The proposed transformation combines two complementary levels of information. First, the triplet $(t, \ell, x_{\ell,t})$ preserves explicit signal-level structure: time position, lead identity, and amplitude. Second, the appended embedding $\mathbf{e}_\ell$ provides a compact lead-wise latent descriptor learned from the autoencoder. In this sense, the classifier receives both local waveform information and a compressed summary of cross-lead structure.

This construction is intuitively appealing, but it also has important limitations. Because the 32-dimensional embedding of a lead is repeated identically across all 1000 time samples of that lead, the latent information is not temporally localized. In addition, the shared MLP used by PointNet treats points independently before global pooling, which means that temporal continuity and local neighborhoods are not explicitly modeled. This is a substantial difference from standard sequence models or convolutional architectures, which are naturally adapted to temporal signals.

A second limitation is that the present implementation does not include the full ablation study requested in the assessment. The code evaluates only the fused representation combining ECG coordinates and embeddings. As a result, there is no direct quantitative comparison between a 2D ECG-only representation and the 2D+embedding representation. The current experiments cannot isolate how much of the final performance comes from the raw ECG structure itself versus the added latent features.

5.6 Discussion

Overall, Part 3 demonstrates that it is feasible to reinterpret a 12-lead ECG and its U-Net-derived embeddings as a structured point set and to classify this representation with a PointNet-inspired model. The transformation is transparent: every original ECG sample becomes one point, augmented with the latent descriptor of its lead. This satisfies the conceptual objective of combining temporal ECG structure with embeddings from Part 2.

However, the present results do not support a claim of superior predictive performance. The PointNet-inspired classifier underperformed both supervised baselines from Part 1, particularly in terms of macro-F1 and pathological-class recall. The normal class remained the easiest to recognize, whereas pathological superclasses were more frequently missed. This pattern is consistent with both class imbalance and the limited temporal inductive bias of the current architecture.

The most defensible conclusion is therefore that the proposed 3D point-set strategy is an interesting representational experiment rather than a demonstrated improvement over standard ECG classifiers.

6 Conclusion

This work studied ECG classification on PTB-XL through supervised baselines, reconstruction-based representation learning, and a PointNet-inspired point-set classifier. The supervised CNN1D and ResNet1D baselines remained the strongest models in terms of classification performance. The U-Net autoencoder, however, learned clinically meaningful cross-lead representations and showed that random-input-lead training improves robustness across source leads. The PointNet-inspired approach demonstrated the feasibility of combining ECG structure and learned embeddings in a unified point-set representation, but did not yet surpass simpler 1D baselines. Overall, the study highlights both the promise and the current limits of representation-driven ECG modeling.

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